

AlloyGraph: Knowledge-Graph-Grounded Multi-Agent Superalloy Design

Open data & knowledge

Nickel Institute handbook

106 alloy records,
open datasheets

OWL 2 DL ontology

RDF graph (GraphDB)
vector index (Weaviate)

Physics-informed ML

XGBoost + Random Forest
50-80 metallurgical features

Multi-agent triangulation

CrewAI · Llama 3.3-70B

ML

physics

KG

Analyst + Reviewer

cross-check three anchors

deterministic guards

Designer agent

inverse design

Stratified evaluation $n = 88$

evaluated by compositional novelty

NEAR

MID

FAR

distance to nearest training alloy

■ KG anchoring: exact retrieval

-30% yield-strength error, known alloys

■ agent triangulation, novel alloys

-18% yield-strength error

■ physics: elastic modulus

-36% elastic-modulus error

13 of 88 'held-out' alloys were
vendor-renamed near-duplicates

Fully open: data · code · models · erratum ledger · seeded evaluation harness